

Airlines Social Listening: From 2015 Complaints to Real-Time Sentiment Intelligence

A two-part social listening analysis of six major US airlines on Twitter/X — starting from a historical complaint dataset (Feb 2015) and extending into a live sentiment-and-topic monitoring model (Jan-Jun 2026) tracking volume, engagement, and what people are actually complaining about.

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Tool: Power BI · DAX

Scope: Feb 2015 & Jan-Jun 2026

Role: Data Analyst

Sources: Kaggle Twitter Airline Sentiment + Social Listening Model

Original model & theme design

116K

TOTAL TWEETS (2026 MODEL)

34.6%

POSITIVE RATE

30.8%

NEGATIVE RATE

2.41

ENGAGEMENT RATE

01 · Overview

Two datasets, one question: what actually drives airline sentiment?

This project combines two distinct data sources on purpose. The first is the well-known public Kaggle "Twitter US Airline Sentiment" dataset — 14.64K real tweets from February 2015, each hand-labeled with a sentiment and, when negative, a specific complaint reason. The second is a broader social-listening model I built covering January–June 2026: 116K tweets across six carriers (American, Delta, JetBlue, Southwest, Spirit, United), tracking not just sentiment but engagement (likes, retweets, replies) and topic-level themes like Delays, Baggage, and Customer Service.

Used together, the 2015 dataset answers "*what specifically do people complain about?*" at the tweet level, while the 2026 model answers "*how is sentiment and engagement trending, and which topics drive it, over time and by airline?*" — two layers of the same underlying question.

RECONCILING THE NUMBERS

The two datasets are not comparable head-to-head: 14.64K tweets from a single month in 2015 vs. 116K tweets across six months in 2026, from different collection methods and airline rosters (the 2015 set includes US Airways and Virgin America, both since merged/discontinued). They're presented side by side as two independent views, not blended into one total.

WHAT SURPRISED ME

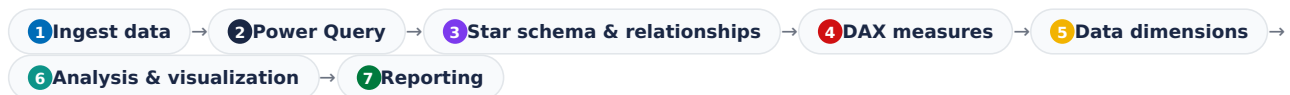
The 2026 model's overall Net Sentiment Score is a modest 3.9% — a number that on its own looks unremarkable. But that single average hides a roughly 29-point swing across airlines: Delta sits at an estimated +13% net sentiment while Spirit sits at around -16%. The aggregate number is technically correct and almost completely uninformative on its own.

Key takeaway: Never report sentiment as one company-wide number when the airline-level spread is this wide — the average erases the story.

02 · Approach

The analyst workflow behind this report

This project follows a standard data-analyst pipeline end to end:



1. Ingest — the public Kaggle Twitter US Airline Sentiment dataset (Feb 2015) and a second social-listening dataset covering six airlines, Jan–Jun 2026, with tweet-level engagement metrics and topic tagging.

2. Power Query — no CSV ingestion — all six tables are authored directly as hardcoded, inline `#table()` constructors. This is a fully synthetic dataset built to demonstrate the model structure, not a cleaned real-world export. Full table-by-table detail is in Section 03.

3. Star schema & relationships — two fact tables, **FactTweets** and **FactTopics**, both related to shared dimensions **DimAirline**, **DimDate**, plus **DimSentiment** (for FactTweets) and **DimTopic** (for FactTopics) — all active many-to-one relationships. Full diagram in Section 03.

4. DAX measures — 14 measures in **FactTweets** (volume, engagement, and sentiment-rate calculations) and 5 in **FactTopics** (topic volume and sentiment) — full list in Section 03.

5. Data dimensions — **DimDate** (MonthLabel, MonthNum, MonthShort, Quarter, SortOrder) drives the Jan–Jun trend views.

6. Analysis & visualization — sentiment trend, net sentiment by airline, engagement, and topic-volume/topic-sentiment views — covered in Section 04.

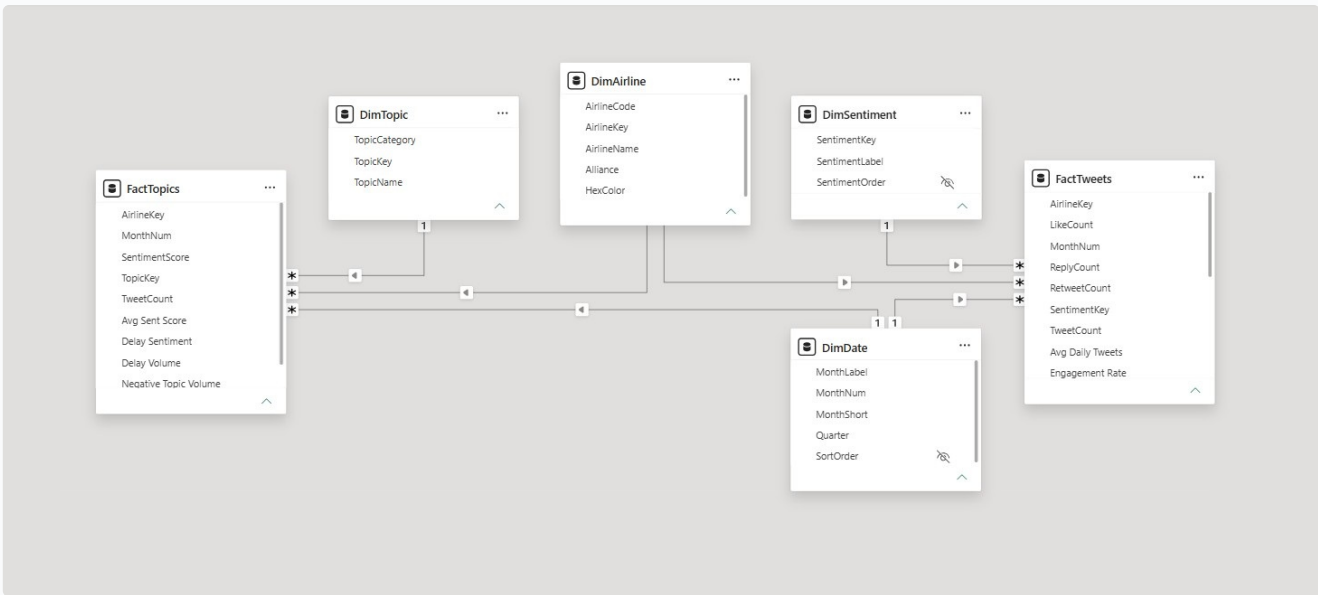
7. Reporting — this document, built from both the historical and live models.

The semantic model and this report's design system were designed and built by me as part of this analysis — not a default template.

03 · Data Model

Semantic model & relationships

The 2026 social-listening model is a star schema with two fact tables sharing common dimensions: **FactTweets** (AirlineKey, MonthNum, SentimentKey, LikeCount, ReplyCount, RetweetCount, TweetCount, plus the *Avg Daily Tweets* and *Engagement Rate* measures) and **FactTopics** (AirlineKey, MonthNum, TopicKey, SentimentScore, TweetCount, plus *Avg Sent Score*, *Delay Sentiment*, *Delay Volume*, and *Negative Topic Volume* measures). Both connect to **DimAirline** (AirlineCode, AirlineName, Alliance, HexColor) and **DimDate** (MonthLabel, MonthNum, MonthShort, Quarter); FactTweets additionally relates to **DimSentiment**, and FactTopics to **DimTopic** (TopicCategory, TopicName).



Semantic model — two fact tables (FactTweets, FactTopics) sharing DimAirline and DimDate, with dedicated DimSentiment and DimTopic dimensions.

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status
☐ FactTopics (AirlineKey)		DimAirline (AirlineKey)	Active ...
☐ FactTopics (MonthNum)		DimDate (MonthNum)	Active ...
☐ FactTopics (TopicKey)		DimTopic (TopicKey)	Active ...
☐ FactTweets (AirlineKey)		DimAirline (AirlineKey)	Active ...
☐ FactTweets (MonthNum)		DimDate (MonthNum)	Active ...
☐ FactTweets (SentimentKey)		DimSentiment (SentimentKey)	Active ...

Manage Relationships view — six active one-to-many relationships, all keyed on AirlineKey, MonthNum, TopicKey, or SentimentKey.

POWER QUERY: A FULLY SYNTHETIC, HARDCODED MODEL

Unlike the LEGO case study — which loaded real CSV exports — this model's six tables are built entirely from inline `#table()` constructors in Power Query. There's no source file and no cleaning step; the data is generated directly to a fixed shape (6 months × 6 airlines × 3 sentiments, and separately × 8 topics) to demonstrate the model design.

Table	Rows	Columns	Notes
DimAirline	6	AirlineKey , AirlineCode , AirlineName , Alliance , HexColor	AA/DL/UA/WN/B6/NK with brand hex colors
DimDate	6	MonthNum , MonthLabel , MonthShort , Quarter , SortOrder	Jan-Jun 2026; label columns sort by SortOrder

DimSentiment	3	SentimentKey , SentimentLabel , SentimentOrder	Positive/Neutral/Negative, sorted by SentimentOrder
DimTopic	8	TopicKey , TopicName , TopicCategory	Delays/Svc/Baggage/Loyalty/Pricing/Comfort/F&B/Safety
FactTweets	108	MonthNum , AirlineKey , SentimentKey , TweetCount , LikeCount , RetweetCount , ReplyCount	6 months × 6 airlines × 3 sentiments
FactTopics	288	MonthNum , AirlineKey , TopicKey , TweetCount , SentimentScore	6 months × 6 airlines × 8 topics

Both fact tables share **DimDate** and **DimAirline**, so a single month or airline slicer filters both simultaneously — confirmed by the six active relationships in the Manage Relationships view above.

KEY DAX MEASURES — FACTTWEETS (14 OF 14)

Total Tweets	#,0
Total Tweets = SUM(FactTweets[TweetCount])	
The base row count for every volume metric in the model.	
Total Likes	#,0
Total Likes = SUM(FactTweets[LikeCount])	
One of the three raw engagement sums feeding Total Engagements below.	
Total Retweets	#,0
Total Retweets = SUM(FactTweets[RetweetCount])	
Second of the three raw engagement sums.	
Total Replies	#,0
Total Replies = SUM(FactTweets[ReplyCount])	
Third of the three raw engagement sums.	
Total Engagements	#,0
Total Engagements = [Total Likes] + [Total Retweets] + [Total Replies]	
Sums the three engagement types into one number — feeds the 279K Total Engagements KPI.	
Engagement Rate	0.00
Engagement Rate = DIVIDE([Total Engagements], [Total Tweets])	
Engagements per tweet — DIVIDE instead of / avoids a divide-by-zero error when a filter context has no tweets. This is what shows Delta out-engaging every other airline despite lower volume.	
Avg Daily Tweets	#,0
Avg Daily Tweets = DIVIDE([Total Tweets], 30)	
A simplified daily average (fixed 30-day divisor) rather than a true day-count — worth knowing if a selected period isn't exactly a month.	
Positive Tweets	#,0
Positive Tweets = CALCULATE([Total Tweets], DimSentiment[SentimentLabel] = "Positive")	
Filters the base measure to one sentiment label rather than filtering the fact table directly — keeps the logic reusable across Positive/Negative/Neutral variants.	

Negative Tweets

#,0

```
Negative Tweets =  
    CALCULATE(  
        [Total Tweets],  
        DimSentiment[SentimentLabel] = "Negative"  
    )
```

Same pattern as **Positive Tweets**, filtered to "Negative" — the two together drive the Net Sentiment Score below.

Neutral Tweets

#,0

```
Neutral Tweets =  
    CALCULATE(  
        [Total Tweets],  
        DimSentiment[SentimentLabel] = "Neutral"  
    )
```

Completes the three-way sentiment split — Positive, Negative, and Neutral Tweets always sum back to Total Tweets.

Positive Rate

0.0%

```
Positive Rate = DIVIDE([Positive Tweets], [Total Tweets])
```

Positive Tweets as a share of all tweets — the 34.6% headline KPI.

Negative Rate

0.0%

```
Negative Rate = DIVIDE([Negative Tweets], [Total Tweets])
```

Mirrors Positive Rate — the 30.8% headline KPI.

Neutral Rate

0.0%

```
Neutral Rate = DIVIDE([Neutral Tweets], [Total Tweets])
```

Completes the three sentiment rates so Positive + Negative + Neutral always sums to 100%.

Net Sentiment Score

0.0%

```
Net Sentiment Score = DIVIDE([Positive Tweets] - [Negative Tweets], [Total Tweets])
```

The single number the Overview callout warns against reading in isolation — 3.9% overall, but ranging from roughly -16% (Spirit) to +13% (Delta) by airline.

KEY DAX MEASURES — FACTTOPICS (5 OF 5)

Topic Volume

#,0

```
Topic Volume = SUM(FactTopics[TweetCount])
```

The base volume measure for every topic-level chart, including the 115K Topic Mentions KPI.

Avg Sent Score

0.00

```
Avg Sent Score = AVERAGE(FactTopics[SentimentScore])
```

A simple row-level average rather than a weighted one — each tagged mention counts equally regardless of engagement.

Delay Volume

#,0

```
Delay Volume =  
    CALCULATE(  
        [Topic Volume],  
        DimTopic[TopicName] = "Delays"  
    )
```

Isolates one topic from the shared **Topic Volume** measure — the 25K Delay Mentions KPI.

Delay Sentiment

0.00

```
Delay Sentiment =  
    CALCULATE(  
        [Avg Sent Score],  
        DimTopic[TopicName] = "Delays"  
    )
```

Same filter pattern applied to **Avg Sent Score** instead of volume — the -0.40 Delay Sentiment KPI, and the most negative number on the whole dashboard.

Negative Topic Volume

#,0

```
Negative Topic Volume =  
    CALCULATE(  
        [Topic Volume],  
        FactTopics[SentimentScore] < 0  
    )
```

Filters on the row-level score directly rather than a named topic — this is a value filter (any topic scoring negative), not a category filter like the two Delay measures above.

ONE MORE MEASURE — LIVES ON THE COLORS TABLE, NOT IN _MEASURES

Color code

COLORS TABLE

```
Color code = IF(HASONEVALUE(colors[rgb]), "#" & MIN(colors[rgb]))
```

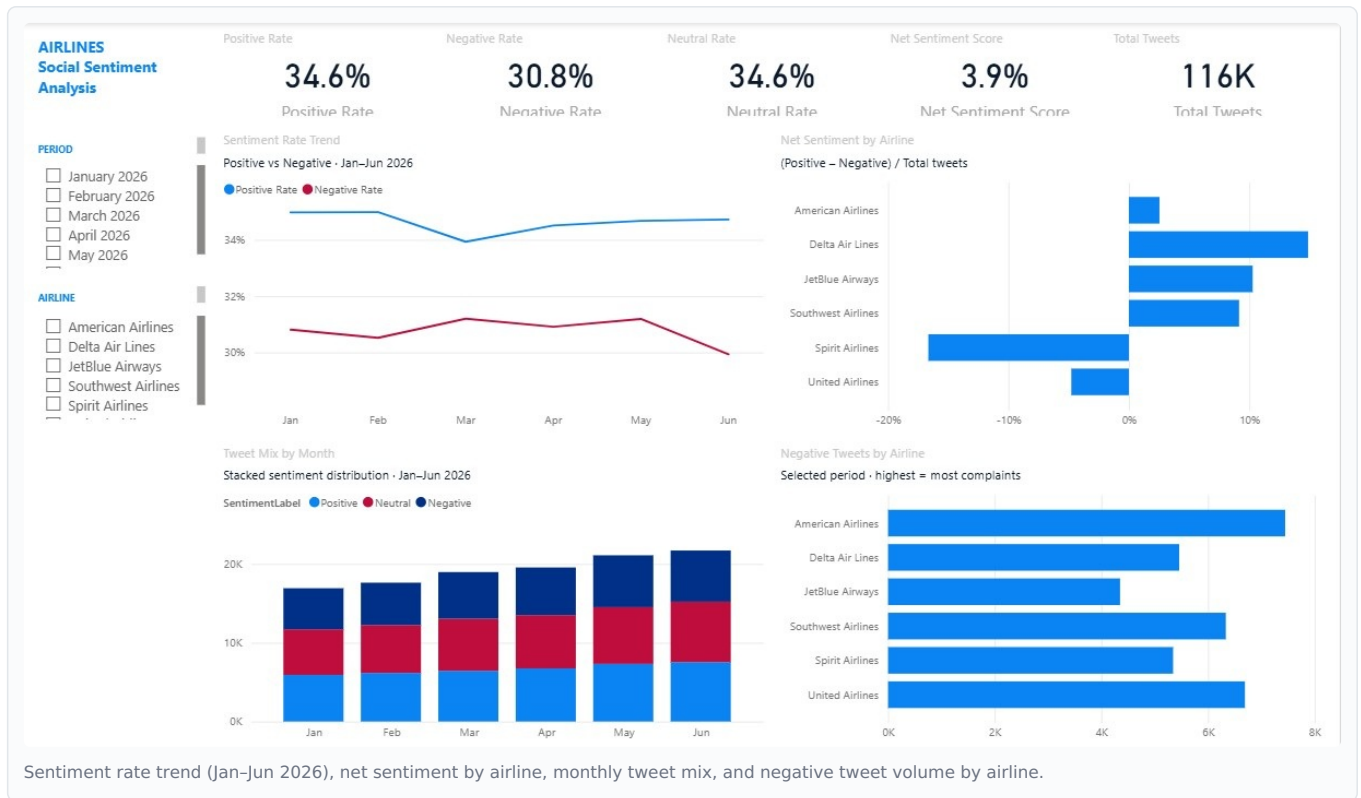
HASONEVALUE guards against an ambiguous result when a visual shows multiple colors at once — it only returns a hex string when exactly one `rgb` value is in context, otherwise blank. Kept on its source table rather than centralized in `_Measures`, since it's tightly coupled to that one column.

04 • Insights

What the two datasets show

2026 MODEL: OVERALL PERFORMANCE SNAPSHOT

Across Jan–Jun 2026, the six airlines generated 116K tweets, 34.6% positive and 30.8% negative (34.6% neutral), for a net sentiment score of just 3.9%. Positive rate trended slightly up and negative rate slightly down over the period, while total tweet mix grew steadily from ~17K to ~22K per month.



NET SENTIMENT SPLITS SHARPLY BY AIRLINE

Delta and JetBlue post the strongest net sentiment (both solidly positive), Southwest and American sit modestly positive, United is slightly negative, and Spirit is the clear outlier at roughly -16% net sentiment — the widest gap in the dataset.



VOLUME AND ENGAGEMENT DON'T TRACK SENTIMENT

American Airlines generates the most tweet volume but only modestly positive sentiment. Delta, despite lower volume than American or Southwest, has both the best net sentiment *and* the highest engagement rate (~2.4+ per tweet). Spirit has the lowest volume and the worst sentiment — a small, vocal, unhappy audience rather than a large one.

AIRLINES Social Sentiment Analysis

PERIOD

- ☐ January 2026
- ☐ February 2026
- ☐ March 2026
- ☐ April 2026
- ☐ May 2026
- ☐ June 2026

Total Tweets

116K

Total Tweets

Avg Daily Tweets

3,862

Avg Daily Tweets

Total Engagements

279K

Total Engagements

Engagement Rate

2.41

Engagement Rate

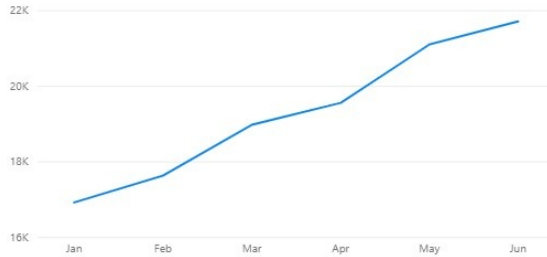
Positive Rate

34.6%

Positive Rate

Volume Trend

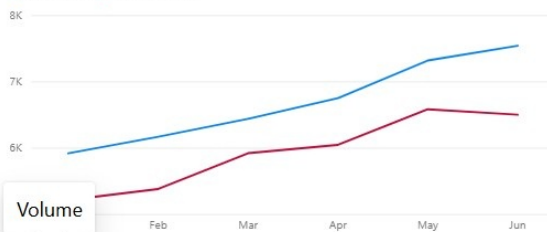
Total tweets per month - Jan-Jun 2026



Positive vs Negative Trend

Monthly tweet volume by sentiment polarity

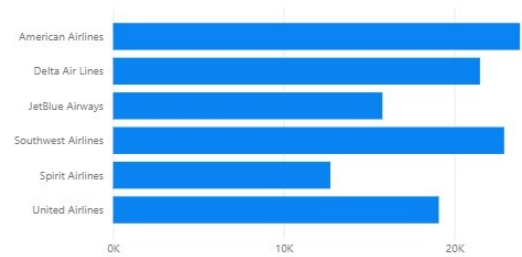
Positive Tweets Negative Tweets



Volume

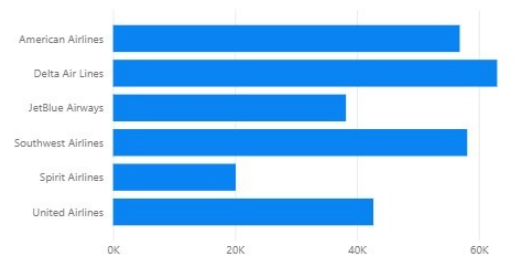
Volume by Airline

Selected period



Engagement by Airline

Likes + retweets + replies - selected period



116K total tweets, 279K total engagements, 2.41 engagement rate — with volume and engagement broken out by airline.

TOPIC ANALYSIS: DELAYS ARE THE PROBLEM, LOYALTY IS THE BRIGHT SPOT

Delays are both the highest-volume topic (~28K mentions) and the most negative (around -0.4 average sentiment) — the single largest driver of negative conversation. Baggage is a smaller but also clearly negative topic. On the positive side, Loyalty/Miles carries the strongest positive sentiment (around +0.3) of any topic, followed by Food & Beverage. Safety has the lowest volume and near-neutral sentiment — barely discussed, in either direction.

AIRLINES Social Sentiment Analysis

AIRLINE

- ☐ American Airlines
- ☐ Delta Air Lines
- ☐ JetBlue Airways
- ☐ Southwest Airlines
- ☐ Spirit Airlines

PERIOD

- ☐ January 2026
- ☐ February 2026
- ☐ March 2026
- ☐ April 2026
- ☐ May 2026

Topic Mentions

115K

Topic Volume

Delay Mentions

25K

Delay Volume

Delay Sentiment

-0.40

Delay Sentiment

Avg Sentiment

-0.03

Avg Sent Score

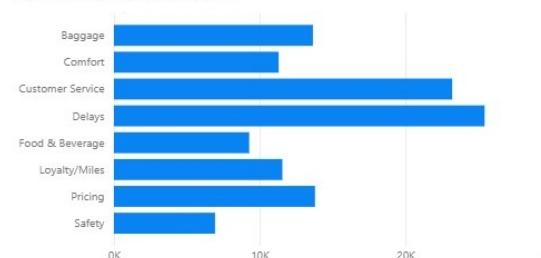
Neg Topic Mentions

62K

Neg Topic Volume

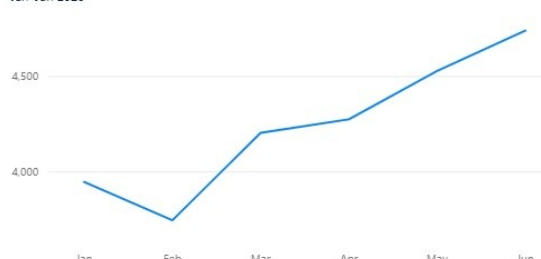
Topic Volume Ranking

Total mentions per topic - selected period



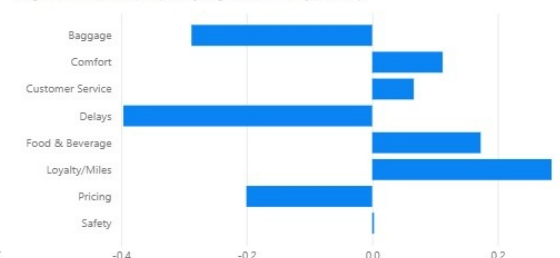
Delay Mentions Trend

Jan-Jun 2026



Topic Sentiment Score

Avg sentiment per topic (-1 very negative to +1 very positive)



Topic Mix by Airline

Stacked - all topics - selected period



Topic volume ranking and topic sentiment score (-1 to +1) — Delays dominate volume and negativity; Loyalty/Miles is the clearest positive outlier.

HISTORICAL BASELINE: THE 2015 COMPLAINT ANATOMY

The Kaggle dataset (14.64K tweets, Feb 2015) skews heavily negative overall — 63% negative vs. 16% positive, with a total sentiment score of -7K. United had the highest raw tweet volume (26.11% of the total), but US Airways posted the worst average sentiment score of any carrier, with American close behind. Virgin America, despite the smallest volume, had by far the least negative average sentiment.

Social Listening Twitter Airlines Complaints

14.64K

Total Tweets

0.16

Percent Positive

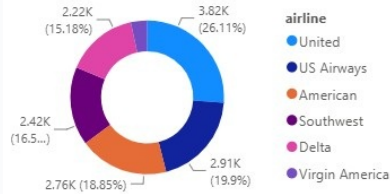
0.63

Percent Negative

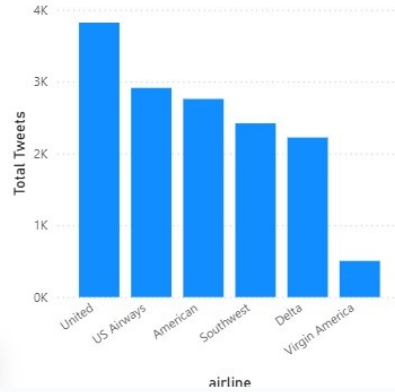
-7K

Sum of SentimentScore

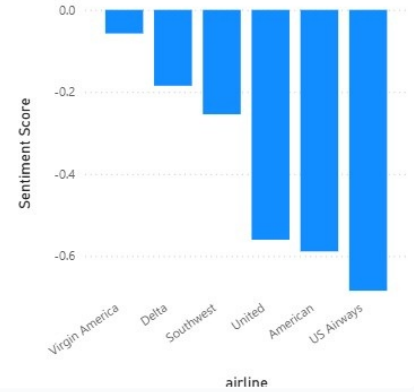
Total Tweets by airline



Total Tweets by airline



Sentiment Score by airline



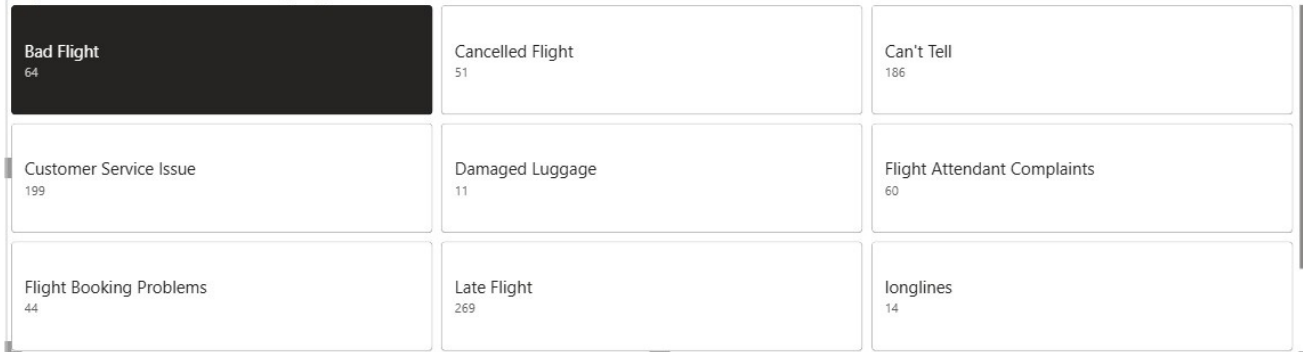
Overview

Source: Public Twitter Data, Kaggle, 2/1/2015–2/28/2015. 14.64K tweets, 63% negative overall.

ROOT CAUSE: WHY PEOPLE COMPLAINED IN 2015

Across the full 2015 dataset, "Customer Service Issue" and "Late Flight" are consistently the top two named complaint reasons, well ahead of categories like Damaged Luggage or Long Lines. Drilling into United specifically (the highest-volume airline in the dataset, 2,633 negative tweets), the same pattern holds: Customer Service Issue (681) and Late Flight (525) dominate, followed by "Can't Tell" (379, meaning the reason wasn't clearly stated) and Lost Luggage (269).

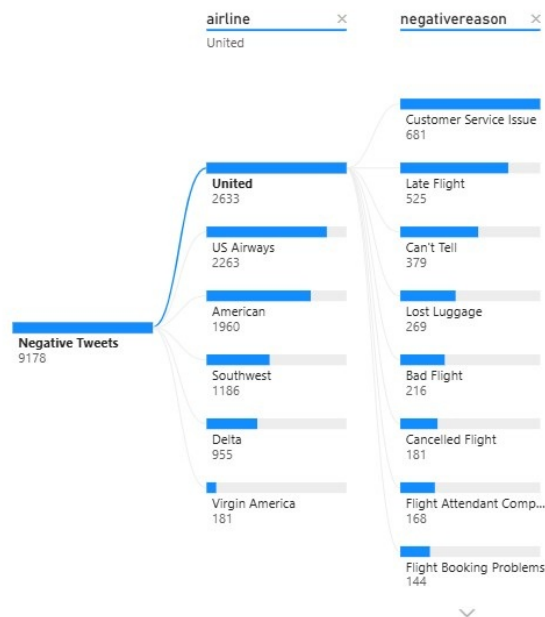
Negative Tweets and First airline by negativereason



Negative Tweets negativereason text

Negative Tweets	negativereason	text
1	Bad Flight	@Jetblue I will keep on tweeting until your CEO answers and personally apologizes to my wife and the passengers of flight 136. Just Wrong!!
1	Bad Flight	@JetBlue if only the show was on the flight. No ABC. Oh well. I'll live. #nexttime #andthewinneris...
1	Bad Flight	@JetBlue #669 stewardesses can't work w/o a galley light! Back to NYC to change a light. #fromthefrontseat #jetblueBlues @anku @yaffasolin
1	Bad Flight	@JetBlue :-/ he was trying to take stuff from the under the seat in front of him and bugging out throughout the flight. Didn't feel safe.
1	Bad Flight	@JetBlue @amybruni @DIRECTTV No Oscars on Direct TV on this flight. We were so looking forward to it. We were devastated. :-/
1	Bad Flight	@JetBlue @amybruni No wifi on this flight so we can't tweet you our Oscar party pics at 37,000ft. #bluecarpet :-/ SEA ✈ BOS
1	Bad Flight	@JetBlue 2nd time flying jetblue this year and my armrest TV controller doesn't work again!!and that's the main reason I fly jetblue 🙄🙄🙄
1	Bad Flight	@JetBlue 653. Just for update that gate got moved to complete opposite end of the terminal. Airline is a disaster.
1	Bad Flight	@JetBlue a voucher.. I can't believe I took this risk. Not only was the plane terribly filthy and I had 15 screaming babies in my face..
1	Bad Flight	@JetBlue also not have food available even for purchase is quite shocking since this is a 5 hour international flight.
1	Bad Flight	@JetBlue Apparently the pilot had made some announcement about engine trouble and losing fuel too quickly.

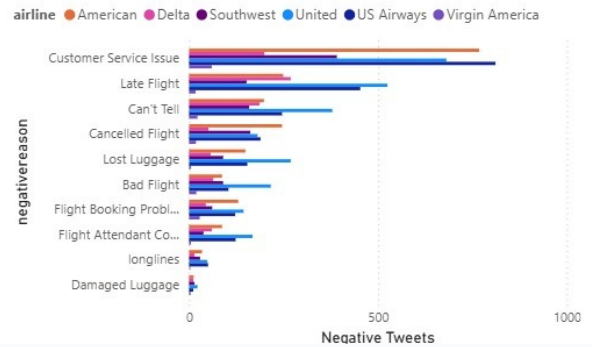
Named complaint reasons across the full dataset — Late Flight and Customer Service Issue lead.



Negative Tweets by negativereason



Negative Tweets by negativereason and airline



9,178 total negative tweets across all six 2015 airlines; United's own breakdown mirrors the aggregate pattern (Customer Service Issue and Late Flight lead).

The overlap is the real finding: eleven years apart, on two different datasets and airline rosters, "customer service" and "delays/late flights" are still the two biggest named drivers of negative sentiment — the 2026 topic model's "Delays" finding is the direct descendant of 2015's "Late Flight" complaint category.

05 • Recommendations

The transferable takeaway

- **Report sentiment per airline, never as one blended number** — a 3.9% overall net sentiment score hides a ~29-point spread between Delta and Spirit; an aggregate KPI alone would miss the story entirely.
- **Delays deserve top operational priority** — highest topic volume and most negative sentiment simultaneously, in the 2026 model, with "Late Flight" already the #2 named complaint reason back in 2015. This is an 11-year-old, structural issue, not a seasonal blip.
- **Loyalty/Miles is an underused positive lever** — the single most positively-sentimented topic; airlines could lean into loyalty messaging specifically where sentiment is already working in their favor.
- **Volume is not a proxy for brand health** — American has the most tweet volume but only modest sentiment; Delta, with less volume, has both better sentiment and higher engagement. Track net sentiment and engagement rate alongside volume, not instead of it.
- **Watch small-but-loud segments** — Spirit's low tweet volume combined with its worst-in-class net sentiment is a concentration-risk pattern worth monitoring, similar to how US Airways showed the worst sentiment in the 2015 baseline despite not having the highest volume.